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Information processing device, signal correction method, and computer-readable recording medium storing signal correction program

Abstract

An information processing device includes a connection circuit to which a device is coupled, an arithmetic processing circuit communicating with the device coupled to the connection circuit through a first communication path, a correction circuit that corrects a signal in the first communication path, and a control circuit. The control circuit receives, from the device, a connection signal indicating that the device is coupled to the connection circuit through a second communication path when the device is coupled to the connection circuit, obtains, from the device, identification information that indicates the device through a third communication path based on the connection signal, updates first correction information to be used to correct the signal to second correction information based on the identification information, and transmits, to the arithmetic processing circuit, an update completion signal through a fourth communication path after the first correction information is updated to the second correction information.

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References Cited**U.S. PATENT DOCUMENTS**

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2006/0161714	12/2005	Satoshi	N/A	N/A
2010/0094928	12/2009	Nakamura	709/203	H04L 67/125
2012/0021780	12/2011	Okuda	455/458	H04M 1/663
2013/0124737	12/2012	Tachibana	709/227	H04L 12/00
2016/0165084	12/2015	Ozawa	358/1.13	H04N 1/00888
2018/0359173	12/2017	Tomita	N/A	H04L 45/02
2019/0005627	12/2018	Nanaumi	N/A	G06T 7/75
2022/0191661	12/2021	Imai	N/A	H04L 45/12

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2006-201881	12/2005	JP	N/A
2009-282798	12/2008	JP	N/A

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Background/Summary**CROSS-REFERENCE TO RELATED APPLICATION**

(1) This application is based upon and claims the benefit of priority of the prior Japanese Patent Application No. 2022-121093, filed on Jul. 29, 2022, the entire contents of which are incorporated herein by reference.

FIELD

(2) The embodiment discussed herein is related to a signal correction technique.

BACKGROUND

(3) With development of software technology, there are increasing cases of constructing a calculation system having high performance and high availability by coupling a large number of

servers via a communication network.

(4) In a case of constructing such a calculation system, it is preferable to keep the unit price and the installation area per server lower while securing the processing performance of each server to some extent. Moreover, in order to construct the calculation system on the communication network, each server may preferably be equipped with a plurality of input/output (I/O) cards of different types.

(5) In relation to the I/O card, there has been known a technique of extending a transmission distance of a Peripheral Component Interconnect (PCI) Express interface. There has also been known an information processing device that takes an appropriate measure when the number of lanes of the PCI Express interface is reduced.

(6) Japanese Laid-open Patent Publication No. 2009-282798 and Japanese Laid-open Patent Publication No. 2006-201881 are disclosed as related art.

SUMMARY

(7) According to an aspect of the embodiments, an information processing device including: a connection circuit to which a device is coupled; an arithmetic processing circuit that communicates with the device coupled to the connection circuit through a first communication path; a correction circuit that corrects a signal in the first communication path; and a control circuit that receives, from the device, a connection signal that indicates that the device is coupled to the connection circuit through a second communication path when the device is coupled to the connection circuit, obtains, from the device, identification information that indicates the device through a third communication path based on the connection signal, updates first correction information to be used to correct the signal to second correction information based on the identification information, and transmits, to the arithmetic processing circuit, an update completion signal through a fourth communication path after the first correction information is updated to the second correction information, wherein the first communication path includes a communication path between the connection circuit and the arithmetic processing circuit, the second communication path includes a communication path between the connection circuit and the control circuit, the third communication path includes a communication path between the connection circuit and the control circuit, and the fourth communication path includes a communication path between the control circuit and the arithmetic processing circuit.

(8) The object and advantages of the invention will be realized and attained by means of the elements and combinations particularly pointed out in the claims.

(9) It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory and are not restrictive of the invention.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a configuration diagram of a server according to a comparative example;

(2) FIG. 2 is a functional configuration diagram of an information processing device according to an embodiment;

(3) FIG. 3 is a flowchart of a signal correction process;

(4) FIG. 4 is a functional configuration diagram illustrating a specific example of the information processing device;

(5) FIG. 5 is a diagram illustrating signal correction data;

(6) FIG. 6 is a diagram illustrating device information;

(7) FIG. 7 is a diagram illustrating setting information;

(8) FIG. 8 is a diagram illustrating a sideband signal;

(9) FIG. 9 is a flowchart of a power supply control process;

(10) FIG. 10 is a flowchart of a control process;

- (11) FIG. 11 is a flowchart of a first event process;
- (12) FIG. 12 is a flowchart of a second event process;
- (13) FIG. 13 is a flowchart of a correction parameter update process;
- (14) FIG. 14 is a flowchart of a communication initialization process;
- (15) FIG. 15 is a flowchart of a correction parameter setting process;
- (16) FIG. 16 is a flowchart of a command execution process;
- (17) FIG. 17 is a flowchart of a first operation;
- (18) FIG. 18 is a flowchart of a second operation; and
- (19) FIG. 19 is a hardware configuration diagram of the information processing device.

DESCRIPTION OF EMBODIMENTS

(20) In the server described above, a plurality of types of I/O cards may preferably be mounted so that various customer requests may be handled. Thus, a connection unit for mounting a detachable I/O card is provided.

(21) In this case, a communication path for high-speed communication is commonly provided between the connection unit and a central processing unit (CPU). The CPU may be called a processor at times. For example, Peripheral Component Interconnect Express (PCIe) has been known as a communication standard of high-speed communication.

(22) A higher communication speed between the CPU and the I/O card is more preferable. In view of the above, in order to improve the communication speed, a scheme in which signal amplitude is smaller and a signal frequency is higher tends to be selected in the communication standard. When the amplitude is made smaller and the frequency is made higher, the influence of noise, which does not raise a problem in a low-speed communication standard, increases, and signal distortion occurs due to signal attenuation, interference from another circuit, or physical characteristics of a motherboard.

(23) In common communication standards as well, signal distortion in high-speed communication is taken into consideration. For example, in the PCIe standard, an attempt is made to improve a signal waveform by correcting signals at a time of signal transmission and at a time of signal reception.

(24) For example, a transmission-side circuit makes correction for emphasizing the rising part and the falling part of the signal, which makes it possible to bring the signal received by a reception-side circuit close to a rectangular wave. A plurality of correction parameters for the rising part and the falling part is prepared in advance, and any of the correction parameters is selected and used for signal correction.

(25) However, in a case where an I/O card different from the I/O card that has been coupled to the connection unit of the server is coupled to the connection unit, signal distortion in the communication path for high-speed communication may not be appropriately corrected with the correction parameters prepared in advance. This makes it difficult to use the different I/O card that has been coupled.

(26) Note that such a problem occurs not only in a case where the I/O card is coupled to the server but also in a case where various devices are coupled to the information processing device. Furthermore, such a problem occurs not only in a case where the CPU and the I/O card communicate with each other but also in a case where various arithmetic processing units in the information processing device and the devices communicate with each other.

(27) In one aspect, an object of the embodiment is, even when a different device is coupled to an information processing device, to enable communication between an arithmetic processing unit and the coupled device.

(28) Hereinafter, an embodiment will be described in detail with reference to the drawings.

(29) FIG. 1 illustrates an exemplary configuration of a server according to a comparative example. A server **601** in FIG. 1 includes a baseboard management controller (BMC) **611**, a CPU **612**, a signal correction circuit **613**, a read only memory (ROM) **614**, and device slots **615-1** to **615-3**.

(30) The BMC **611** manages the server **601**. The device slots **615-i** ($i=1$ to 3) correspond to a connection unit, and a detachable device such as an I/O card is coupled to the device slots **615-i**. The ROM **614** stores correction parameters for devices coupled to the device slots **615-i** in association with the individual device slots **615-i**. Throughout this specification, the term “unit” is used as a synonym of a “circuit block”.

(31) The CPU **612** communicates with the devices coupled to the individual device slots **615-i** through a communication path **621-1** and a communication path **621-2** for high-speed communication. The communication path **621-1** and the communication path **621-2** represent communication paths to each of the device slots **615-1** to **615-3**. (In FIG. 1, a connection between the communication path **621-2** and the slot **615-1** is illustrated, and connections between the communication path **621-2** and other slots **615-2** and **615-3** are omitted to be illustrated.)

(32) The signal correction circuit **613** is provided between the communication path **621-1** and the communication path **621-2**, and corrects signals transmitted and received between the CPU **612** and the individual devices using the correction parameters stored in the ROM **614**.

(33) When the main power supply of the server **601** is turned on, the CPU **612** checks sideband signals transmitted from the individual devices. The sideband signals include information indicating a device mounting status and a light emitting diode (LED) status.

(34) When it is determined that a device is mounted based on the sideband signal, the CPU **612** carries out incorporation of the device. In the device incorporation, the device is subject to initialization of high-speed communication. When the initialization of high-speed communication is complete, the CPU **612** checks whether or not link-up of the device is successful.

(35) When the link-up is successful, the CPU **612** obtains vendor information and type information from the device through the communication path **621-1** and the communication path **621-2**. Then, the CPU **612** checks whether or not the correction parameters associated with the individual device slots **615-i** in the ROM **614** correspond to the vendor information and the type information of the devices coupled to the device slots **615-i**.

(36) When the correction parameter associated with the device slot **615-i** does not correspond to the vendor information and the type information of the device coupled to the device slot **615-i**, the CPU **612** instructs the BMC **611** to rewrite the ROM **614**.

(37) The BMC **611** rewrites the ROM **614** through a communication path **622** for low-speed communication in accordance with the instruction from the CPU **612**. As a result, the correction parameters associated with the individual device slots **615-i** in the ROM **614** are updated to correction parameters corresponding to the vendor information and the type information of the devices coupled to the device slots **615-i**.

(38) Next, the BMC **611** restarts the server **601**. At the time of the restart, the correction parameters associated with the individual device slots **615-i** in the ROM **614** correspond to the vendor information and the type information of the devices coupled to the device slots **615-i**. The signal correction circuit **613** obtains the correction parameters for the individual devices from the ROM **614** through the communication path **622**.

(39) The CPU **612** transmits signals directed to the individual devices to the signal correction circuit **613**. The signal correction circuit **613** corrects the signals received from the CPU **612** using the correction parameters for the individual devices, thereby generating correction signals. Then, the signal correction circuit **613** transmits the correction signals to the individual devices.

(40) The individual devices transmit signals directed to the CPU **612** to the signal correction circuit **613**. The signal correction circuit **613** corrects the signals received from the individual devices using the correction parameters for the individual devices, thereby generating correction signals. Then, the signal correction circuit **613** transmits the correction signals to the CPU **612**.

(41) However, when a new device that appears after the release of the server **601** is coupled to the device slot **615-i**, a compatibility problem may occur between the server **601** and the new device, and link-up may fail. When the link-up fails, the CPU **612** is not notified of the vendor information

and the type information of the new device, whereby the ROM **614** is not rewritten and the correction parameters in the ROM **614** are not updated.

(42) When the correction parameters in the ROM **614** are not updated, signal correction using the correction parameter for the new device is not carried out in the signal correction circuit **613**, whereby it is difficult to use the new device.

(43) FIG. **2** illustrates an exemplary functional configuration of an information processing device according to the embodiment. An information processing device **701** in FIG. **2** includes an arithmetic processing unit **711**, a correction unit **712**, a connection unit **713**, and a control unit **714**. The arithmetic processing unit **711** communicates with a device coupled to the connection unit **713** through a first communication path. The correction unit **712** corrects signals in the first communication path.

(44) FIG. **3** is a flowchart illustrating an exemplary signal correction process to be performed by the information processing device **701** in FIG. **2**. First, when a device is coupled to the connection unit **713**, the control unit **714** receives, from the device, a connection signal indicating that the device is coupled to the connection unit **713** through a second communication path (step **801**). Then, the control unit **714** obtains, from the device, identification information indicating the device through a third communication path based on the connection signal (step **802**).

(45) Next, the control unit **714** updates first correction information for signal correction to second correction information based on the identification information (step **803**), and transmits an update completion signal to the arithmetic processing unit **711** through a fourth communication path (step **804**).

(46) The first communication path is a communication path between the connection unit **713** and the arithmetic processing unit **711**, the second communication path is a communication path between the connection unit **713** and the control unit **714**, the third communication path is a communication path between the connection unit **713** and the control unit **714**, and the fourth communication path is a communication path between the control unit **714** and the arithmetic processing unit **711**.

(47) According to the information processing device **701** in FIG. **2**, even in a case where a different device is coupled to the information processing device **701**, it becomes possible to perform communication between the arithmetic processing unit **711** and the coupled device.

(48) FIG. **4** illustrates a specific example of the information processing device **701** in FIG. **2**. An information processing device **901** in FIG. **4** is, for example, a server, and includes a control device **911**, a host unit **912**, a power supply control unit **913**, a correction unit **914**, a storage unit **915**, device slots **916-1** to **916-3**, and a selection unit **917**.

(49) The control device **911** includes a control unit **921**, a storage unit **922**, and communication units **923** to **927**. The host unit **912** includes a communication unit **931**, a communication unit **932**, an arithmetic processing unit **933**, a communication unit **934**, and a storage unit **935**.

(50) The arithmetic processing unit **933**, the correction unit **914**, the device slots **916-i** ($i=1$ to 3), and the control unit **921** correspond to the arithmetic processing unit **711**, the correction unit **712**, the connection unit **713**, and the control unit **714** in FIG. **2**, respectively. The number of the device slots **916-i** is not limited to three, and may be N (N is an integer of 1 or more).

(51) A detachable device such as an I/O card is coupled to the device slot **916-i**. The device to be coupled to the device slot **916-i** may be a network interface card (NIC), a graphics processing unit (GPU), or a solid state drive (SSD).

(52) The control device **911** and the storage unit **915** operate by permanent power supply **P1**, and the host unit **912** operates by normal power supply **P2**. A device coupled to the device slot **916-i** operates by the permanent power supply **P1** and the normal power supply **P2**. When an alternating current (AC) cable is coupled to the information processing device **901**, power of the permanent power supply **P1** is supplied.

(53) The power supply control unit **913** switches ON and OFF of the normal power supply **P2** in

accordance with an instruction from the control device **911**. Power of the normal power supply P2 is supplied when the normal power supply P2 is ON, and the power supply of the normal power supply P2 stops when the normal power supply P2 is OFF.

(54) The storage unit **915** stores signal correction data. The signal correction data includes correction parameters associated with the individual device slots **916-i**. The correction parameters associated with the individual device slots **916-i** indicate correction parameters to be applied to devices coupled to the individual device slots **916-i**. The storage unit **915** is an exemplary correction information storage unit, and the correction parameters are exemplary correction information.

(55) FIG. 5 illustrates an example of the signal correction data stored in the storage unit **915**. The signal correction data in FIG. 5 includes a port number and a correction parameter. The port number and the correction parameter are associated with each other.

(56) The port number is identification information indicating a port for high-speed communication between the host unit **912** and the devices coupled to the individual device slots **916-i**. In the high-speed communication, a plurality of ports is used in the high-speed communication between the host unit **912** and each device. The correction parameter is associated with the device slot **916-i** to which the device using the port indicated by the port number is coupled via the port number.

(57) Returning to FIG. 4, the host unit **912** communicates with the devices coupled to the individual device slots **916-i** through a communication path **941-1** and a communication path **941-2** for high-speed communication. The communication path **941-1** and the communication path **941-2** represent communication paths to each of the device slots **916-1** to **916-3**.

(58) The correction unit **914** is provided between the communication path **941-1** and the communication path **941-2**, and corrects signals transmitted and received between the host unit **912** and the individual devices using the correction parameters stored in the storage unit **915**. The communication path **941-1** and the communication path **941-2** correspond to the first communication path. (In FIG. 4, connections between the communication paths **941-2**, **942**, etc. and the slot **916-1** are illustrated, and connections between those communication paths and other slots **916-2** and **916-3** are omitted to be illustrated.)

(59) The selection unit **917** selects any of the device slots **916-i** in accordance with an instruction from the control device **911**. Then, it transfers the vendor information and the type information output from the device coupled to the selected device slot **916-i** to the control device **911**. The vendor information is identification information indicating a vendor of the device, and the type information is identification information indicating the device.

(60) The arithmetic processing unit **933** of the host unit **912** performs various types of information processing using the storage unit **935**. The arithmetic processing unit **933** transmits, to the communication path **941-1**, signals directed to the device coupled to the device slot **916-i** via the communication unit **934**, and receives signals transmitted from the device from the communication path **941-1**. The arithmetic processing unit **933** communicates with the control device **911** via the communication unit **932**.

(61) The control unit **921** of the control device **911** performs processing of managing the information processing device **901** using the storage unit **922**. The control unit **921** receives an instruction from a user via the communication unit **923**. The instruction from the user is, for example, an instruction for switching ON and OFF of the normal power supply P2, which is input to the information processing device **901** via a web user interface (UI), for example. The control unit **921** communicates with the host unit **912** via the communication unit **925**.

(62) The control device **911** communicates with the correction unit **914**, the storage unit **915**, and the selection unit **917** through a communication path **943** for low-speed communication. In the low-speed communication, the control unit **921** communicates with the correction unit **914**, the storage unit **915**, and the selection unit **917** via the communication unit **926**. The communication path **943** corresponds to the third communication path. The control unit **921** communicates with the

power supply control unit **913** via the communication unit **927**.

(63) The device coupled to the device slot **916-i** initializes the communication function for low-speed communication when the power of the permanent power supply **P1** is supplied. This enables low-speed communication between the device and the control device **911** through the communication path **943**.

(64) When the power of the normal power supply **P2** is supplied, the device starts to stand by for initialization of high-speed communication by the host unit **912**. When the initialization of high-speed communication is carried out, high-speed communication between the device and the host unit **912** becomes available through the communication path **941-1** and the communication path **941-2**.

(65) The storage unit **922** stores power supply information, device information, and setting information. The power supply information indicates whether the normal power supply **P2** is ON or OFF. The device information includes the vendor information and the type information obtained from the devices coupled to the individual device slots **916-i**. The setting information includes the correction parameters for the individual devices, and is set by the user in advance. The storage unit **922** is an exemplary correspondence relationship storage unit.

(66) FIG. **6** illustrates an example of the device information stored in the storage unit **922**. The device information in FIG. **6** includes a slot number, vendor information, and type information. The slot number, the vendor information, and the type information are associated with each other. The slot number is identification information indicating the device slot **916-i** to which the device is coupled.

(67) FIG. **7** illustrates an example of the setting information stored in the storage unit **922**. The setting information in FIG. **7** includes a slot number, vendor information, type information, a port number, a correction parameter, and an update command. The slot number, the vendor information, the type information, the port number, the correction parameter, and the update command are associated with each other.

(68) Each slot number is associated with a plurality of port numbers. Those port numbers indicate a plurality of ports to be used in the high-speed communication with the device coupled to the device slot **916-i** indicated by the slot number.

(69) The correction parameter represents a correction parameter to be applied to the device indicated by the vendor information and the type information when the device is coupled to the device slot **916-i** indicated by the slot number. The update command represents a command to be used when the control device **911** directly updates the correction parameter set in the correction unit **914**. The update command includes each port number corresponding to the slot number and the correction parameter to be applied to the device.

(70) "Other" in the vendor information, the type information, and the port number indicates one or a plurality of general-purpose devices, and the vendor information, the type information, and the port number other than that indicates a specific device. A specific device is, for example, a device to be subject to extreme signal correction.

(71) In the signal correction data stored in the storage unit **915**, a correction parameter associated with each port number is any of the correction parameters included in the setting information.

(72) The devices coupled to the individual device slots **916-i** transmit sideband signals directed to the host unit **912** to the control device **911** through a communication path **942**, and the control unit **921** receives the sideband signals from the devices via a communication unit **924**. The communication path **942** corresponds to the second communication path.

(73) The control unit **921** transmits the received sideband signals to the host unit **912** via the communication unit **924** and a communication path **944**, and the arithmetic processing unit **933** receives the sideband signals from the control device **911** via the communication unit **931**. The communication path **944** corresponds to the fourth communication path. The arithmetic processing unit **933** transmits, to the device, the sideband signal directed to the device via the communication

unit **931**.

(74) FIG. **8** illustrates an example of the sideband signal in high-speed communication based on PCIe. The sideband signal in FIG. **8** include *Atnled_*, *Pwrled*, *pwren_n*, *button_n*, *prsnt_n*, *pwrflt_n*, *mrl_n*, and *Emil*. *Atnled_*, *Pwrled*, *pwren_n*, and *Emil* are information to be notified from the host unit **912** to the device, and *button_n*, *prsnt_n*, *pwrflt_n*, and *mrl_n* are information to be notified from the device to the host unit **912**.

(75) Among the pieces of information to be notified from the device to the host unit **912**, *prsnt_n* is a negative logic device present signal, and indicates whether or not the device is coupled to the corresponding device slot **916-i**. When the device is coupled to the device slot **916-i**, *prsnt_n* is Low, and when no device is coupled to the device slot **916-i**, *prsnt_n* is High. The Low status of *prsnt_n* corresponds to the connection signal indicating that the device is coupled to the connection unit.

(76) Hereinafter, a state in which a device is coupled to the device slot **916-i** may be referred to as a device mounted state, and a state in which no device is coupled to the device slot **916-i** may be referred to as a device non-mounted state.

(77) When the sideband signal received from the device changes from the device mounted state to the device non-mounted state, the control unit **921** of the control device **911** obtains the vendor information and the type information from the device via the communication unit **926**. Then, the control unit **921** stores, in the storage unit **922**, the device information including the slot number corresponding to the sideband signal and the obtained vendor information and type information.

(78) Next, the control unit **921** obtains, from the setting information stored in the storage unit **922**, the port number, the correction parameter, and the update command corresponding to the slot number, the vendor information, and the type information included in the device information.

(79) Next, the control unit **921** refers to the power supply information stored in the storage unit **922** to check whether the normal power supply P2 is ON or OFF. In a case where the normal power supply P2 is OFF and the power is not supplied to the correction unit **914**, the control unit **921** updates the correction parameters in the storage unit **915** via the communication unit **926**. At this time, the control unit **921** updates the correction parameter associated with each port number obtained from the setting information in the storage unit **915** to the correction parameter obtained from the setting information.

(80) In this case, the correction parameter associated with each port number in the storage unit **915** corresponds to the first correction information, and the correction parameter obtained from the setting information corresponds to the second correction information.

(81) Thereafter, when the normal power supply P2 changes from OFF to ON and the power is supplied to the correction unit **914**, the correction unit **914** obtains, from the storage unit **915**, the correction parameter associated with each port number through the communication path **943**.

(82) After updating the correction parameter in the storage unit **915**, the control unit **921** transmits the sideband signal received from the device to the host unit **912** via the communication unit **924** and the communication path **944**. When the arithmetic processing unit **933** of the host unit **912** receives the sideband signal from the control device **911**, it starts to communicate with the device that has transmitted the sideband signal via the communication unit **934**. The sideband signal transmitted from the control device **911** to the host unit **912** corresponds to the update completion signal.

(83) Since the sideband signal is transmitted to the host unit **912** after the correction parameter in the storage unit **915** is updated, the link-up does not start using the correction parameter before the update, which avoids link-up failure.

(84) As an example, it is assumed that a device different from the device that has been coupled to the device slot **916-i** is coupled to the device slot **916-i**. In this case, since the storage unit **915** stores the correction parameter for the different device when the power is supplied to the correction unit **914**, the correction unit **914** may obtain the correction parameter from the storage unit **915**. As

a result, appropriate signal correction is carried out using the correction parameter for the different device, whereby high-speed communication between the arithmetic processing unit **933** and the different device becomes available.

(85) On the other hand, in a case where the normal power supply **P2** is ON and the power is supplied to the correction unit **914**, the control unit **921** updates the correction parameter in the storage unit **915** via the communication unit **926**. At this time, the control unit **921** updates the correction parameter associated with each port number obtained from the setting information in the storage unit **915** to the correction parameter obtained from the setting information.

(86) Moreover, the control unit **921** transmits the update command obtained from the setting information to the correction unit **914** via the communication unit **926**. As a result, the control unit **921** updates the correction parameter set in association with each port number corresponding to the slot number in the device information in the correction unit **914** to the correction parameter included in the update command.

(87) In this case, the correction parameter set in association with each port number in the correction unit **914** corresponds to the first correction information, and the correction parameter included in the update command corresponds to the second correction information.

(88) After updating the correction parameter in the storage unit **915** and the correction parameter in the correction unit **914**, the control unit **921** transmits the sideband signal received from the device to the host unit **912** via the communication unit **924**. When the arithmetic processing unit **933** of the host unit **912** receives the sideband signal from the control device **911**, it starts to communicate with the device that has transmitted the sideband signal via the communication unit **934**.

(89) Since the sideband signal is transmitted to the host unit **912** after the correction parameter in the correction unit **914** is updated, the link-up does not start using the correction parameter before the update, which avoids link-up failure.

(90) As an example, in a case where a device different from the device that has been coupled to the device slot **916-i** is coupled to the device slot **916-i**, the correction parameter in the correction unit **914** is updated to the correction parameter for the different device. As a result, appropriate signal correction is carried out using the correction parameter for the different device, whereby high-speed communication between the arithmetic processing unit **933** and the different device becomes available.

(91) According to the information processing device **901** in FIG. **4**, it becomes possible to perform high-speed communication between the arithmetic processing unit **933** and a different device coupled to the device slot **916-i**, whereby the different device may be used without a problem. Even if the different device is a new device that has appeared after the release of the information processing device **901**, a similar effect may be obtained.

(92) Even in a case of using a device to be subject to extreme signal correction, the correction parameter may be automatically set in the storage unit **915** and the correction unit **914** by setting the correction parameter for the device in the setting information in advance.

(93) FIG. **9** is a flowchart illustrating an exemplary power supply control process to be performed by the power supply control unit **913** in FIG. **4**. First, when the power supply control unit **913** receives a control signal from the control device **911**, it checks the received control signal (step **1401**).

(94) If the control signal indicates OFF (YES in step **1401**), the power supply control unit **913** switches the normal power supply **P2** from ON to OFF (step **1402**), and repeats the process of step **1401** and subsequent steps. On the other hand, if the control signal indicates ON (NO in step **1401**), the power supply control unit **913** switches the normal power supply **P2** from OFF to ON (step **1403**), and repeats the process of step **1401** and subsequent steps.

(95) FIG. **10** is a flowchart illustrating an exemplary control process to be performed by the control unit **921** in FIG. **4**. First, the control unit **921** checks whether or not an event has been detected (step **1501**).

(96) If an event has been detected (YES in step **1501**), the control unit **921** performs event processing corresponding to the detected event (step **1502**), and repeats the process of step **1501** and subsequent steps. On the other hand, if no event has been detected (NO in step **1501**), the control unit **921** repeats the process of step **1501** and subsequent steps.

(97) FIG. **11** is a flowchart illustrating exemplary first event processing to be performed in step **1502** when the event detected in step **1501** in FIG. **10** is reception of a power supply control instruction. The control device **911** receives a power supply control instruction for the normal power supply **P2** from the host unit **912** or the user. The power supply control instruction indicates OFF, ON, or a power cycle. The power cycle corresponds to a restart.

(98) First, the control unit **921** checks whether or not the received power supply control instruction indicates OFF (step **1601**). If the power supply control instruction indicates OFF (YES in step **1601**), the control unit **921** transmits a control signal indicating OFF to the power supply control unit **913** via the communication unit **927** (step **1602**). Then, the control unit **921** sets the power supply information stored in the storage unit **922** to OFF (step **1603**).

(99) On the other hand, if the power supply control instruction indicates ON or the power cycle (NO in step **1601**), the control unit **921** checks whether or not the power supply control instruction indicates ON (step **1604**). If the power supply control instruction indicates ON (YES in step **1604**), the control unit **921** transmits a control signal indicating ON to the power supply control unit **913** via the communication unit **927** (step **1605**). Then, the control unit **921** sets the power supply information stored in the storage unit **922** to ON (step **1603**).

(100) On the other hand, if the power supply control instruction indicates the power cycle (NO in step **1604**), the control unit **921** transmits a control signal indicating OFF to the power supply control unit **913** via the communication unit **927** (step **1606**). Next, after waiting for a certain period of time (step **1607**), the control unit **921** transmits a control signal indicating ON to the power supply control unit **913** (step **1608**). Then, the control unit **921** sets the power supply information stored in the storage unit **922** to ON (step **1603**).

(101) FIG. **12** is a flowchart illustrating exemplary second event processing to be performed in step **1502** when the event detected in step **1501** in FIG. **10** is a change of a sideband signal. The control device **911** monitors the sideband signal for each of the device slots **916-i**, and performs the event processing of FIG. **12** when the sideband signal has changed from the sideband signal at the time of initial reception.

(102) First, the control unit **921** refers to the received sideband signal (step **1701**), and checks whether or not the device connection status in the device slot **916-i** has changed (step **1702**).

(103) If the device connection status has not changed (NO in step **1702**), the control unit **921** changes the sideband signal being output to the host unit **912** via the communication unit **924** to the received sideband signal (step **1703**).

(104) On the other hand, if the device connection status has changed (YES in step **1702**), the control unit **921** checks whether or not the device non-mounted state has changed to the device mounted state (step **1704**).

(105) If the device mounted state has changed to the device non-mounted state (NO in step **1704**), the control unit **921** performs the processing of step **1703**. When the device is removed from the device slot **916-i**, the device connection status changes from the device mounted state to the device non-mounted state.

(106) On the other hand, if the device non-mounted state has changed to the device mounted state (YES in step **1704**), the control unit **921** performs correction parameter update processing (step **1705**), and performs the processing of step **1703**. When the device is coupled to the device slot **916-i**, the device connection status changes from the device non-mounted state to the device mounted state.

(107) FIG. **13** is a flowchart illustrating exemplary correction parameter update processing in step **1705** in FIG. **12**. First, the control unit **921** transmits, to the selection unit **917**, a selection

instruction indicating the device slot **916-i** that has changed from the device non-mounted state to the device mounted state via the communication unit **926** (step **1801**). The selection unit **917** selects the device slot **916-i** indicated by the selection instruction received from the control device **911**.

(108) Next, the control unit **921** obtains the vendor information and the type information from the device coupled to the selected device slot **916-i** via the communication unit **926** (step **1802**). Then, the control unit **921** stores, in the storage unit **922**, the device information including the slot number indicating the selected device slot **916-i** and the obtained vendor information and type information (step **1803**).

(109) Next, the control unit **921** obtains, from the storage unit **915**, the port number and the correction parameter via the communication unit **926** (step **1804**). Next, the control unit **921** obtains, from the setting information stored in the storage unit **922**, the port number, the correction parameter, and the update command corresponding to the slot number, the vendor information, and the type information included in the device information (step **1805**).

(110) Then, the control unit **921** compares the correction parameter obtained from the storage unit **915** with the correction parameter obtained from the setting information for each port number obtained from the setting information (step **1806**).

(111) If the correction parameter obtained from the storage unit **915** matches the correction parameter obtained from the setting information for each port number (YES in step **1806**), the control unit **921** ends the process.

(112) On the other hand, if the correction parameter obtained from the storage unit **915** does not match the correction parameter obtained from the setting information for one or a plurality of port numbers obtained from the setting information (NO in step **1806**), the control unit **921** performs processing of step **1807**. In step **1807**, the control unit **921** updates the correction parameter associated with each port number obtained from the setting information in the storage unit **915** to the correction parameter obtained from the setting information via the communication unit **926**.

(113) Next, the control unit **921** refers to the power supply information stored in the storage unit **922** (step **1808**) to check whether the normal power supply P2 is ON or OFF (step **1809**). If the normal power supply P2 is OFF (NO in step **1809**), the control unit **921** ends the process.

(114) On the other hand, if the normal power supply P2 is ON (YES in step **1809**), the control unit **921** transmits, to the correction unit **914**, the update command obtained from the setting information via the communication unit **926** (step **1810**). As a result, the control unit **921** updates the correction parameter set in association with each port number corresponding to the slot number in the device information in the correction unit **914** to the correction parameter included in the update command.

(115) FIG. **14** is a flowchart illustrating an exemplary communication initialization process to be performed by the arithmetic processing unit **933** in FIG. **4**. When the normal power supply P2 is turned ON, the arithmetic processing unit **933** performs the communication initialization process in FIG. **14**.

(116) First, the arithmetic processing unit **933** refers to the sideband signal of each of the device slots **916-i** received from the control device **911** to check whether or not the device connection status is the device mounted state (step **1901**).

(117) If the device connection status is the device mounted state (YES in step **1901**), the arithmetic processing unit **933** initializes the communication with the device coupled to the device slot **916-i** via the communication unit **934** (step **1902**). As a result, the device incorporation is carried out, and high-speed communication between the arithmetic processing unit **933** and the device becomes available.

(118) If the device connection status is the device non-mounted state (NO in step **1901**), the arithmetic processing unit **933** ends the process.

(119) When the communication initialization process in FIG. **14** is terminated, an operating system

(OS) starts, and the arithmetic processing unit **933** starts various types of information processing using the storage unit **935**.

(120) FIG. **15** is a flowchart illustrating an exemplary correction parameter setting process to be performed by the correction unit **914** in FIG. **4**. The correction unit **914** performs the correction parameter setting process in FIG. **15** when the normal power supply **P2** is turned ON.

(121) First, the correction unit **914** obtains, from the storage unit **915**, the correction parameter corresponding to each port number through the communication path **943** (step **2001**). Then, the correction unit **914** sets the obtained correction parameter as a correction parameter corresponding to the port indicated by the port number (step **2002**).

(122) Next, the correction unit **914** checks whether or not a command is received from the control device **911** (step **2003**). The control device **911** transmits, to the correction unit **914**, either an update command or a read command, for example.

(123) If the command is received (YES in step **2003**), the correction unit **914** executes the received command (step **2004**), and repeats the process of step **2003** and subsequent steps. On the other hand, if no command is received (NO in step **2003**), the correction unit **914** repeats the process of step **2003** and subsequent steps.

(124) FIG. **16** is a flowchart illustrating an exemplary command execution process in step **2004** in FIG. **15**. First, the correction unit **914** checks whether or not the received command is an update command (step **2101**).

(125) If the received command is an update command (YES in step **2101**), the correction unit **914** updates the correction parameter set to the port indicated by the port number included in the update command to the correction parameter included in the update command (step **2102**). Then, the correction unit **914** transmits a response indicating the update result to the control device **911** (step **2103**).

(126) On the other hand, if the received command is a read command (NO in step **2101**), the correction unit **914** transmits, to the control device **911**, a response including the correction parameter set to the port indicated by the port number included in the read command (step **2104**).

(127) FIG. **17** is a flowchart illustrating an exemplary first operation when a device is coupled to the device slot **916-i** before the power of the permanent power supply **P1** is supplied to the information processing device **901** in FIG. **4**.

(128) First, the user couples the device to any of the device slots **916-i** (step **2201**), and couples an AC cable to the information processing device **901** (step **2202**). As a result, the power of the permanent power supply **P1** is supplied to the control device **911**, the storage unit **915**, and the device.

(129) The control unit **921** performs the control process in FIG. **10**, and performs the correction parameter update processing in step **1705** in FIG. **12**. Then, the control unit **921** updates the correction parameter associated with each port number obtained from the setting information in the storage unit **915** to the correction parameter obtained from the setting information in step **1807** in FIG. **13** (step **2203**). In this case, the normal power supply **P2** is OFF, and thus the processing of step **1810** is skipped.

(130) Next, the user transmits a power supply control instruction for turning the normal power supply **P2** ON to the control device **911** (step **2204**). In step **1605** in FIG. **11**, the control unit **921** transmits a control signal indicating ON to the power supply control unit **913**, and in step **1403** in FIG. **9**, the power supply control unit **913** switches the normal power supply **P2** from OFF to ON. As a result, the power of the normal power supply **P2** is supplied to the host unit **912**, the correction unit **914**, and the device.

(131) The correction unit **914** performs the correction parameter setting process in FIG. **15**, and in step **2002**, it sets the correction parameter corresponding to each port number obtained from the storage unit **915** as a correction parameter corresponding to the port indicated by the port number (step **2205**). As a result, the correction parameter for the device coupled to the device slot **916-i** is

set in the correction unit **914**.

(132) Next, the arithmetic processing unit **933** performs the communication initialization process in FIG. **14**. Since the correction parameter suitable for the device is set in the correction unit **914**, upon completion of the communication initialization process, the link-up of the device succeeds, and the high-speed communication between the arithmetic processing unit **933** and the device becomes available. Therefore, the information processing device **901** is enabled to perform information processing using the coupled device.

(133) Meanwhile, at a time of active replacement of the device, the device is removed from the device slot **916-i** and another device is coupled to the device slot **916-i** while the power of the permanent power supply **P1** is being supplied.

(134) FIG. **18** is a flowchart illustrating an exemplary second operation when a device is coupled to the device slot **916-i** while the power of the permanent power supply **P1** is being supplied to the information processing device **901** in FIG. **4**.

(135) First, the user couples an AC cable to the information processing device **901** (step **2301**). As a result, the power of the permanent power supply **P1** is supplied to the control device **911**, the storage unit **915**, and the device.

(136) Next, the user transmits a power supply control instruction for turning the normal power supply **P2** ON to the control device **911** (step **2302**). The control unit **921** performs the control process in FIG. **10**, and transmits a control signal indicating ON to the power supply control unit **913** in step **1605** in FIG. **11**, and the power supply control unit **913** switches the normal power supply **P2** from OFF to ON in step **1403** in FIG. **9**. As a result, the power of the normal power supply **P2** is supplied to the host unit **912**, the correction unit **914**, and the device.

(137) The correction unit **914** performs the correction parameter setting process in FIG. **15**, and in step **2002**, it sets the correction parameter corresponding to each port number obtained from the storage unit **915** as a correction parameter corresponding to the port indicated by the port number.

(138) Next, the user couples the device to any of the device slots **916-i** (step **2303**). After removing the device from any of the device slots **916-i**, the user may couple another device to that device slot **916-i**.

(139) The control unit **921** performs the correction parameter update processing in step **1705** in FIG. **12**. Then, the control unit **921** updates the correction parameter associated with each port number obtained from the setting information in the storage unit **915** to the correction parameter obtained from the setting information in step **1807** in FIG. **13** (step **2304**).

(140) In this case, since the normal power supply **P2** is ON, the control unit **921** transmits the update command obtained from the setting information to the correction unit **914** in step **1810** (step **2305**). Then, in step **2102** in FIG. **16**, the correction unit **914** updates the correction parameter set to the port indicated by the port number included in the update command to the correction parameter included in the update command. As a result, the correction parameter for the device coupled to the device slot **916-i** is set in the correction unit **914**.

(141) Next, the arithmetic processing unit **933** performs the communication initialization process in FIG. **14**. Since the correction parameter suitable for the device is set in the correction unit **914**, upon completion of the communication initialization process, the link-up of the device succeeds, and the high-speed communication between the arithmetic processing unit **933** and the device becomes available.

(142) As described above, the information processing device **901** also supports the active replacement of the device, and the link-up of the device succeeds even when the device is coupled to the device slot **916-i** while the power of the permanent power supply **P1** is being supplied. Therefore, the information processing device **901** is enabled to perform information processing using the coupled device.

(143) FIG. **19** illustrates an exemplary hardware configuration of the information processing device **901** in FIG. **4**. An information processing device **2401** in FIG. **19** includes a BMC **2411**, an

interface (IF) **2412**, an IF **2413**, a CPU **2414**, an IF **2415**, and a memory **2416**. The information processing device **2401** further includes a power supply control circuit **2417**, a signal correction circuit **2418**, a ROM **2419**, device slots **2420-1** to **2420-3**, and a multiplexer (MUX) **2421**. Those components are hardware.

(144) The BMC **2411** includes a CPU **2431**, a memory **2432**, and iFs **2433** to **2437**. Those components are hardware.

(145) The BMC **2411** operates as the control device **911** in FIG. 4. The CPU **2431**, the memory **2432**, and the iFs **2433** to **2437** operate as the control unit **921**, the storage unit **922**, and the communication units **923** to **927** in FIG. 4, respectively.

(146) The IF **2412**, the IF **2413**, the CPU **2414**, the IF **2415**, and the memory **2416** operate as the communication unit **931**, the communication unit **932**, the arithmetic processing unit **933**, the communication unit **934**, and the storage unit **935** in FIG. 4, respectively. The power supply control circuit **2417**, the signal correction circuit **2418**, the ROM **2419**, and the MUX **2421** operate as the power supply control unit **913**, the correction unit **914**, the storage unit **915**, and the selection unit **917** in FIG. 4, respectively.

(147) The device slots **2420-i** ($i=1$ to 3) correspond to the device slots **916-i** in FIG. 4. The number of the device slots **2420-i** is not limited to three, and may be N (N is an integer of 1 or more).

(148) The IF **2415** communicates with devices coupled to the individual device slots **2420-i** through a communication path **2441-1** and a communication path **2441-2** for high-speed communication. The IF **2436** of the BMC **2411** communicates with the signal correction circuit **2418**, the ROM **2419**, and the MUX **2421** through a communication path **2443** for low-speed communication.

(149) The IF **2434** of the BMC **2411** communicates with the devices coupled to the individual device slots **2420-i** through a communication path **2442**, and communicates with the IF **2412** through a communication path **2444**.

(150) The communication path **2441-1**, the communication path **2441-2**, the communication path **2442**, the communication path **2443**, and the communication path **2444** correspond to the communication path **941-1**, the communication path **941-2**, the communication path **942**, the communication path **943**, and the communication path **944** in FIG. 4, respectively. (In FIG. 19, connections between the communication paths **2441-2**, **2442**, etc. and the slot **2420-1** are illustrated, and connections between those communication paths and other slots **2420-2** and **2420-3** are omitted to be illustrated.)

(151) The communication path **2441-1** and the communication path **2441-2** are, for example, communication paths based on PCIe, and the communication path **2443** is, for example, a communication path based on the inter-integrated circuit (I2C). The communication path **2442** and the communication path **2444** are, for example, communication paths for PCIe sideband signals. The IF **2415** is, for example, a PCIe interface, and the IF **2413**, the IF **2435**, and the IF **2436** are, for example, I2C interfaces. The device slots **2420-i** are, for example, PCIe slots.

(152) The memory **2416** and the memory **2432** are semiconductor memories such as a random access memory (RAM). The ROM **2419** may be an Electrically Erasable Programmable Read-Only Memory (EEPROM). The IF **2433** is, for example, a local area network (LAN) interface.

(153) The memory **2416** stores a program **X1** and data to be used for information processing. The CPU **2414** operates as the arithmetic processing unit **933** by executing the program **X1** using the memory **2416**.

(154) The memory **2432** stores a program **X2** (firmware) and data to be used for processing of the BMC **2411**. The CPU **2431** operates as the control unit **921** by executing the program **X2** using the memory **2432**.

(155) In a case where the device coupled to the device slot **2420-i** is an SSD, the SSD may store the program **X1** and the program **X2**. In this case, the program **X1** and the program **X2** are loaded from the SSD to the memory **2416** and to the memory **2432**, respectively.

(156) In a case where the device coupled to the device slot **2420-i** is a medium driving device, the medium driving device drives a portable recording medium **2451** to access the recorded content. The portable recording medium **2451** may store the program **X1** and the program **X2**. In this case, the program **X1** and the program **X2** are loaded from the portable recording medium **2451** to the memory **2416** and to the memory **2432**, respectively.

(157) The portable recording medium **2451** is a memory device, a flexible disk, an optical disk, a magneto-optical disk, or the like. The portable recording medium **2451** may be a compact disk read only memory (CD-ROM), a digital versatile disk (DVD), a universal serial bus (USB) memory, or the like.

(158) As described above, the computer-readable recording medium for storing the program **X1** and the program **X2** is a physical (non-transitory) recording medium such as the memory **2416**, the memory **2432**, the SSD, or the portable recording medium **2451**.

(159) The configurations of the information processing device **701** in FIG. 2, the information processing device **901** in FIG. 4, and the information processing device **2401** in FIG. 19 are merely examples, and a part of the components may be omitted or modified depending on the use or conditions of the information processing devices.

(160) The configuration of the server **601** in FIG. 1 is merely an example, and a part of the components may be omitted or modified depending on the use or conditions of the server **601**.

(161) The flowcharts of FIG. 3 and FIGS. 9 to 18 are merely examples, and a part of the processes may be omitted or modified depending on the configuration or conditions of the information processing devices.

(162) The signal correction data illustrated in FIG. 5, the device information illustrated in FIG. 6, the setting information illustrated in FIG. 7, and the sideband signal illustrated in FIG. 8 are merely examples. The signal correction data, the device information, the setting information, and the sideband signal in another format may be used depending on the configuration or conditions of the information processing device **901**.

(163) While the disclosed embodiment and the advantages thereof have been described in detail, those skilled in the art will be able to make various modifications, additions, and omissions without departing from the scope of the embodiment as explicitly set forth in the claims.

(164) All examples and conditional language provided herein are intended for the pedagogical purposes of aiding the reader in understanding the invention and the concepts contributed by the inventor to further the art, and are not to be construed as limitations to such specifically recited examples and conditions, nor does the organization of such examples in the specification relate to a showing of the superiority and inferiority of the invention. Although one or more embodiments of the present invention have been described in detail, it should be understood that the various changes, substitutions, and alterations could be made hereto without departing from the spirit and scope of the invention.

Claims

1. An information processing device comprising: a connection circuit to which a first device is coupled; an arithmetic processing circuit that communicates with the first device coupled to the connection circuit through a first communication path; a correction circuit configured to correct signal distortion of a signal, wherein the signal distortion is caused by high-speed communication of the signal between the arithmetic processing circuit and the first device in the first communication path based on correction information stored in a correction information storage; and a control circuit that receives, from a second device, a connection signal that indicates that the second device is coupled to the connection circuit through a second communication path when the second device is coupled to the connection circuit, obtains, from the second device, identification information that indicates the second device through a third communication path based on the

connection signal, updates first correction information that is stored in the correction information storage and is used to correct the signal distortion for the first device to second correction information that is used for the second device and is obtained from a correspondence relationship storage using the identification information, and transmits, to the arithmetic processing circuit, an update completion signal through a fourth communication path after the first correction information is updated to the second correction information, wherein the first communication path includes a communication path between the connection circuit and the arithmetic processing circuit, the second communication path includes a communication path between the connection circuit and the control circuit, the third communication path includes a communication path between the connection circuit and the control circuit, and the fourth communication path includes a communication path between the control circuit and the arithmetic processing circuit, wherein the control circuit updates the first correction information stored in the correction information storage to the second correction information based on the identification information when power is not supplied to the correction circuit, and the correction circuit obtains the second correction information from the correction information storage when the power is supplied to the correction circuit.

2. The information processing device according to claim 1, wherein the arithmetic processing circuit starts communication between the arithmetic processing circuit and the first device through the first communication path when the update completion signal is received from the control circuit.

3. The information processing device according to claim 1, wherein when power is supplied to the correction circuit, the control circuit updates the first correction information set in the correction circuit to the second correction information based on the identification information.

4. The information processing device according to claim 1, wherein the correspondence relationship storage stores the second correction information in association with the identification information, wherein the control circuit obtains the second correction information from the correspondence relationship storage based on the identification information.

5. A signal correction method to be performed by an information processor including a connection circuit and an arithmetic processing circuit that communicates with a first device coupled to the connection circuit through a first communication path, the method comprising: receiving, when a second device is coupled, a connection signal that indicates that the second device is coupled to the connection circuit through a second communication path, wherein signal distortion of a signal is caused by high-speed communication of the signal between the arithmetic processing circuit and the first device in the first communication path, and is corrected by a correction circuit based on correction information stored in a correction information storage; obtaining identification information that indicates the second device through a third communication path based on the connection signal; updating first correction information that is stored in the correction information storage and is used to correct the signal distortion for the first device to second correction information that is used for the second device and is obtained from a correspondence relationship storage using the identification information; transmitting an update completion signal to the arithmetic processing circuit through a fourth communication path after the first correction information is updated to the second correction information; wherein the first communication path includes a communication path between the connection circuit and the arithmetic processing circuit, the second communication path includes a communication path between the connection circuit and a control circuit, the third communication path includes a communication path between the connection circuit and the control circuit, and the fourth communication path includes a communication path between the control circuit and the arithmetic processing circuit; wherein in the updating; the first correction information stored in the correction information storage is updated to the second correction information based on the identification information when power is not supplied to the correction circuit, and the correction circuit obtains the second correction

information from the correction information storage when the power is supplied to the correction circuit.

6. The signal correction method according to claim 5, wherein the arithmetic processing circuit starts communication between the arithmetic processing circuit and the first device through the first communication path when the update completion signal is received from the control circuit.

7. The signal correction method according to claim 5, wherein in the updating; when power is supplied to the correction circuit, the first correction information set in the correction circuit is updated to the second correction information based on the identification information.

8. The signal correction method according to claim 5, further comprising: obtaining the second correction information from the correspondence relationship storage that stores the second correction information in association with the identification information, based on the identification information.

9. A non-transitory computer-readable recording medium storing a signal correction program causing a processor, that includes a connection circuit and an arithmetic processing circuit that communicates with a first device coupled to the connection circuit through a first communication path, to perform a process comprising: receiving, when a second device is coupled, a connection signal that indicates that the second device is coupled to the connection circuit through a second communication path, wherein signal distortion of a signal is caused by high-speed communication of the signal between the arithmetic processing circuit and the first device in the first communication path, and is corrected by a correction circuit based on correction information stored in a correction information storage; obtaining identification information that indicates the second device through a third communication path based on the connection signal; updating first correction information that is stored in the correction information storage and is used to correct the signal distortion for the first device to second correction information that is used for the second device and is obtained from a correspondence relationship storage using the identification information; transmitting an update completion signal to the arithmetic processing circuit through a fourth communication path after the first correction information is updated to the second correction information; wherein the first communication path includes a communication path between the connection circuit and the arithmetic processing circuit, the second communication path includes a communication path between the connection circuit and a control circuit, the third communication path includes a communication path between the connection circuit and the control circuit, and the fourth communication path includes a communication path between the control circuit and the arithmetic processing circuit; wherein in the updating; the first correction information stored in the correction information storage is updated to the second correction information based on the identification information when power is not supplied to the correction circuit, and the correction circuit obtains the second correction information from the correction information storage when the power is supplied to the correction circuit.

10. The recording medium according to claim 9, wherein the arithmetic processing circuit starts communication between the arithmetic processing circuit and the first device through the first communication path when the update completion signal is received from the control circuit.

11. The recording medium according to claim 9, wherein in the updating; when power is supplied to the correction circuit, the first correction information set in the correction circuit is updated to the second correction information based on the identification information.

12. The recording medium according to claim 9, further comprising: obtaining the second correction information from the correspondence relationship storage that stores the second correction information in association with the identification information, based on the identification information.
